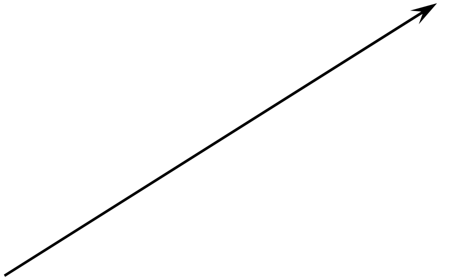
Computer Graphics Basics - I

Contents

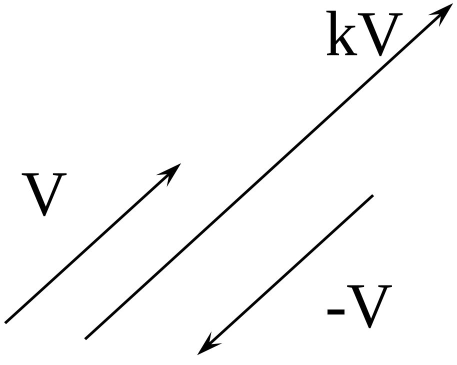
Vector

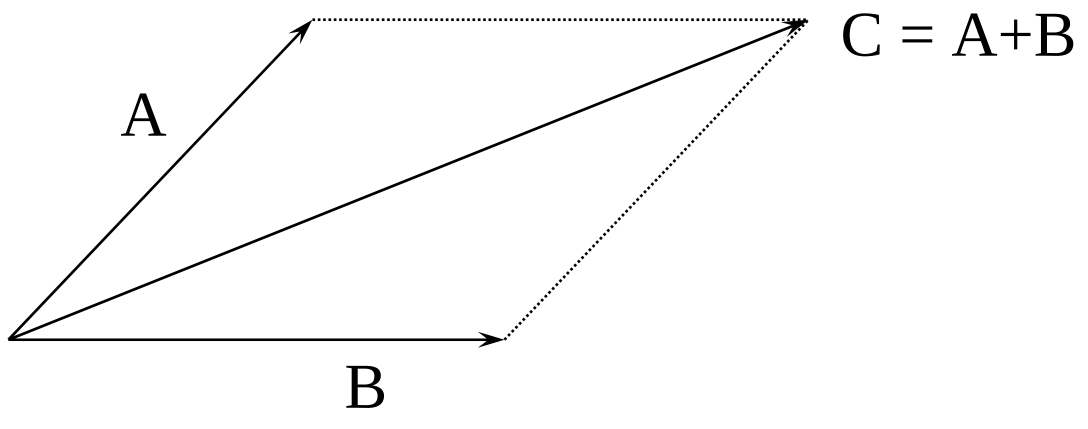
- Vector has length and direction
- Representation
- Usually drawn as segment with arrow-head



Vector Operations

- Scalar Multiplication
- Negative of a Vector
- Addition of Vectors





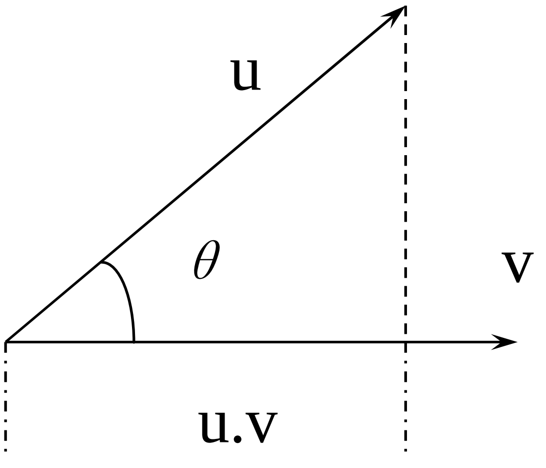
Vector Operations (contd.)

- Dot (Inner) Product

  u.v = *u v* cos(θ)

- Length (Norm) of Vector

  ||v|| = sqrt(v.v)



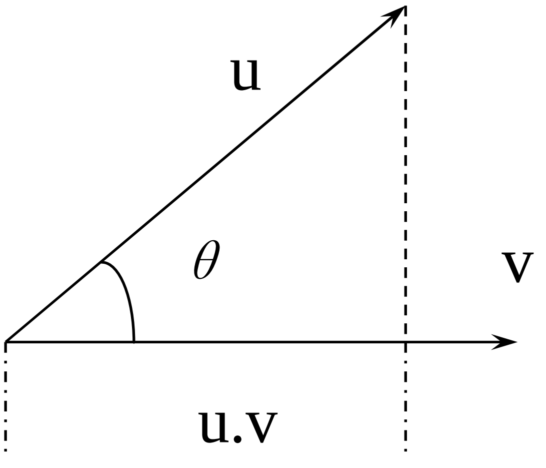
- Normalization of a Vector

$$\mathbf{v}' = \frac{\mathbf{v}}{\left\|\mathbf{v}\right\|} = \frac{1}{\sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}}\mathbf{v}$$

Vector Operations (contd.)

- Dot (Inner) Product

$$\mathbf{u} \cdot \mathbf{v} = \sum_i u_i v_i$$



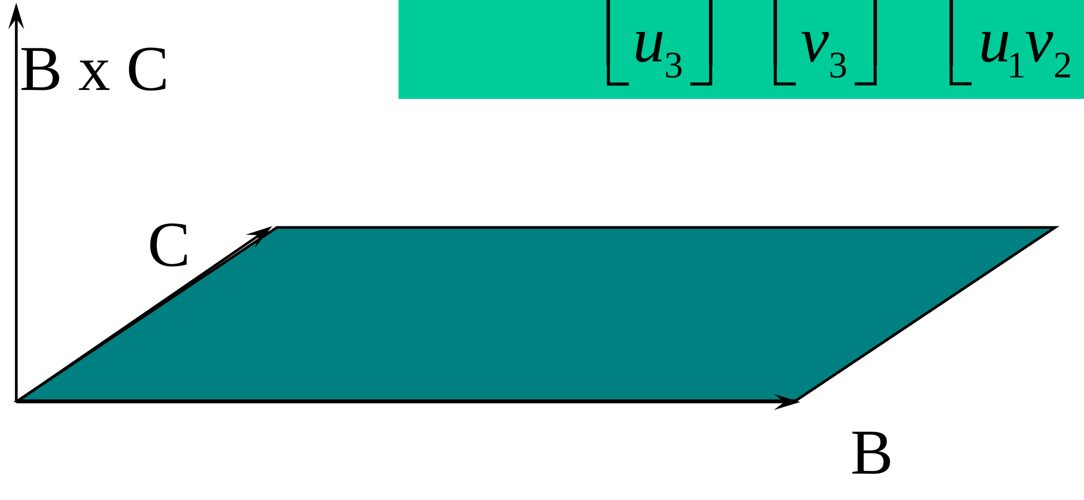
$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos\theta$$

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v} = \begin{bmatrix} u_1 & u_2 & u_3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = u_1 v_1 + u_2 v_2 + u_3 v_3$$

Vector Operations (contd.)

- Cross Product

$$\mathbf{u} \times \mathbf{v} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \times \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{bmatrix}$$



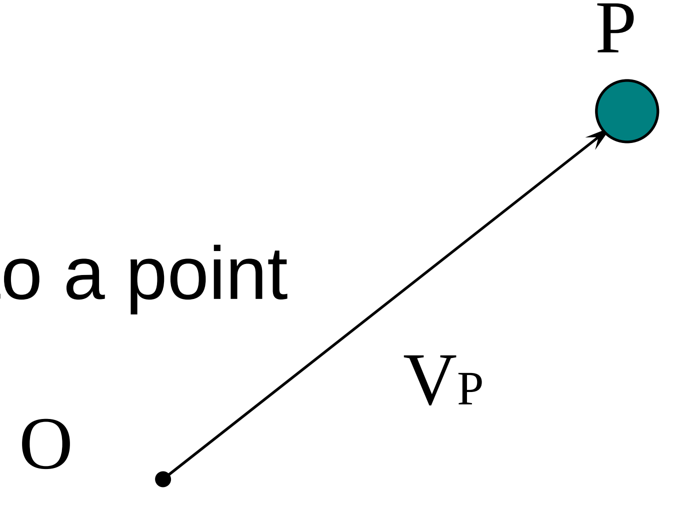
Scalar Triple Product A.(BxC) ?

Point

- Point has a position in space
- Origin is a special point

  O = (0, 0, …)
- Position Vector

  Vector joining origin to a point

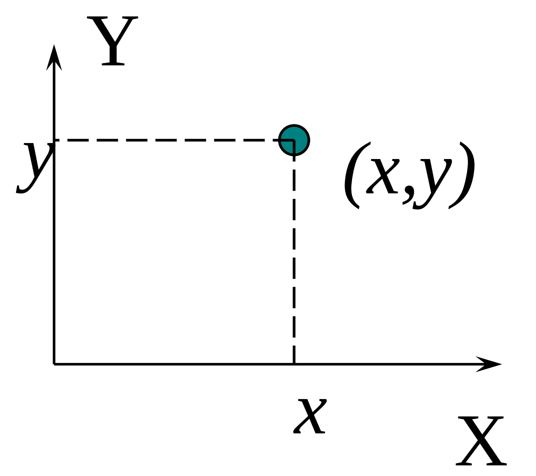


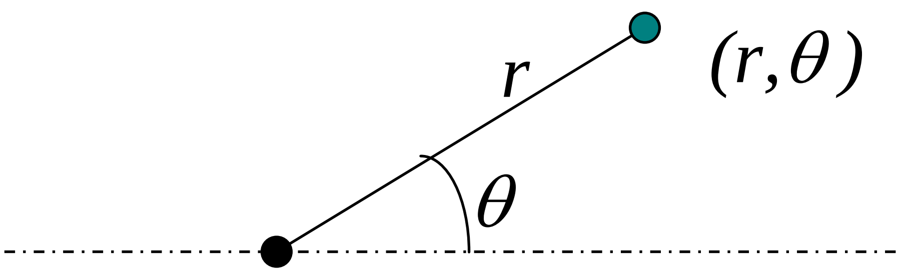
Co-ordinate Systems

- Concept of Dimension

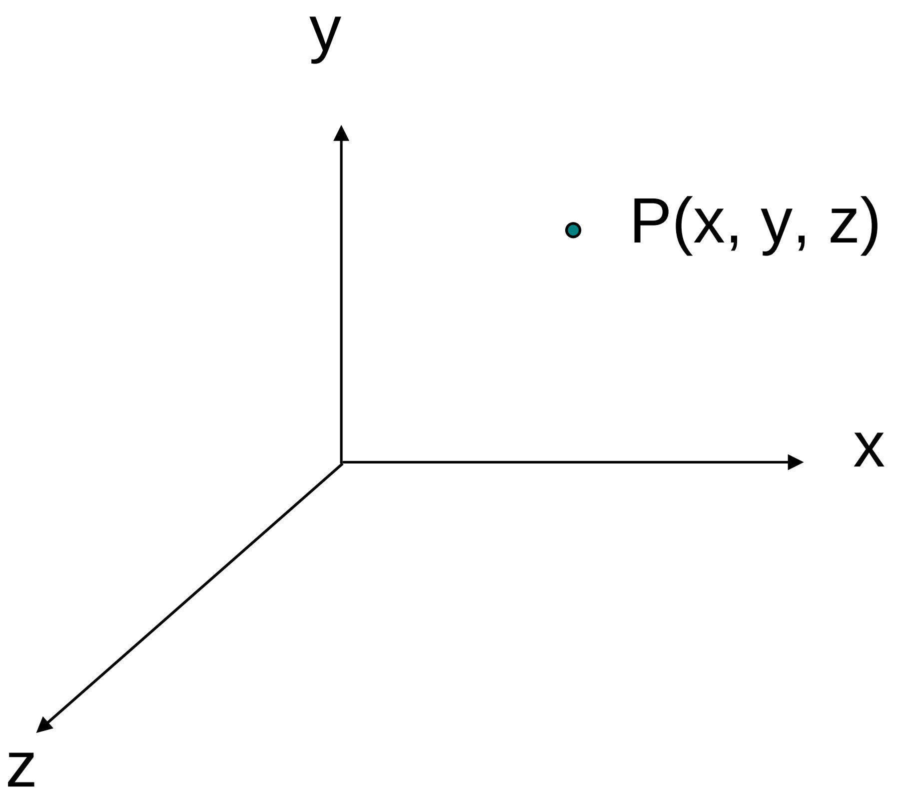
  Dimension of Space v/s Dimension of Object

- Cartesian Co-ordinates
- Polar Co-ordinates



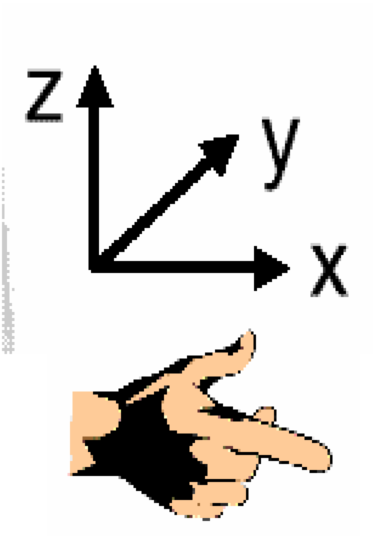


3D Coordinates

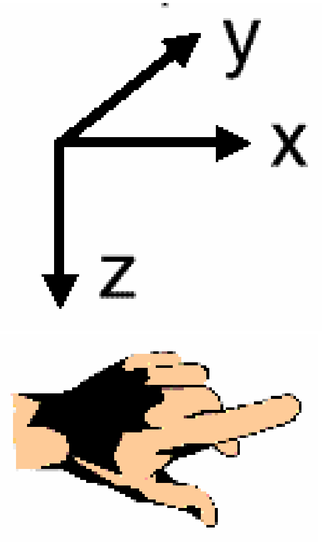


3D Coordinate Systems

- Coordinate systems let us define points in 3D



Right handed



Left handed

Points and Vectors

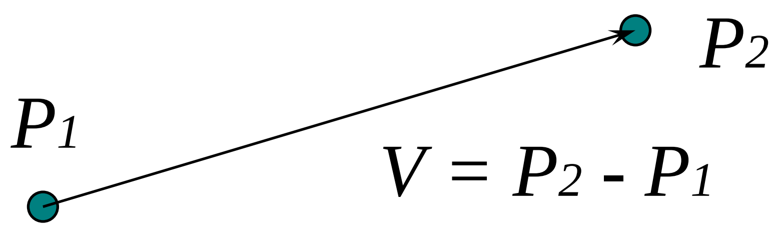
- Are Points different from Vectors?

  Yes!

  Vector operations can not be done on points

- Displacement Vector

  A vector denoting transition (translation) between two points



$$V = P_2 - P_1$$

Line

- Representation

    (Start Point, End Point)

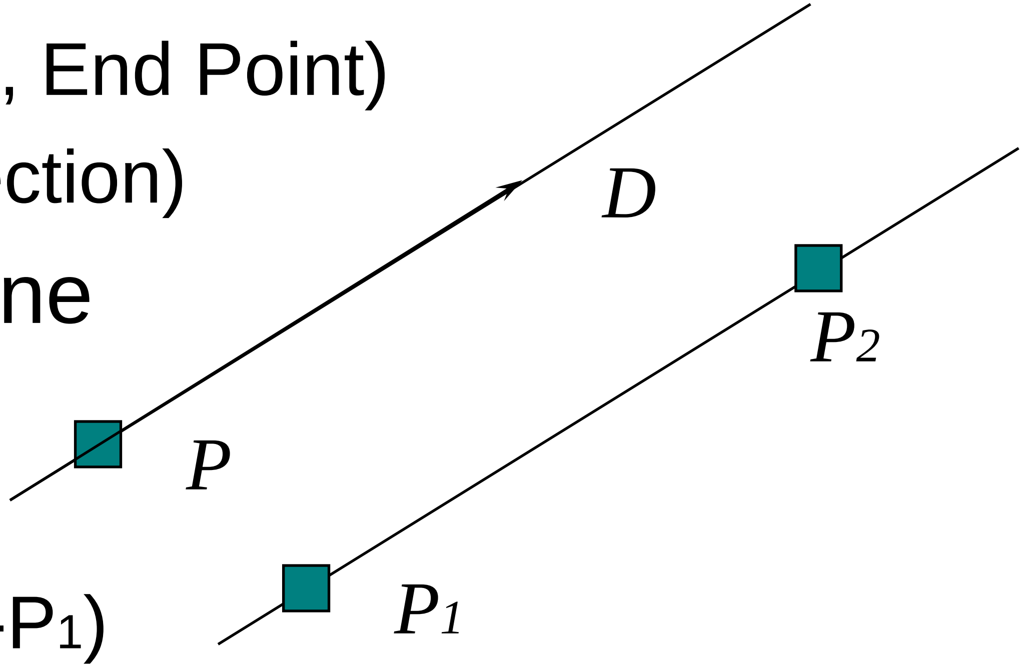
    or (Point, Direction)

- Equation of Line

    $L = P + t * D$

    or

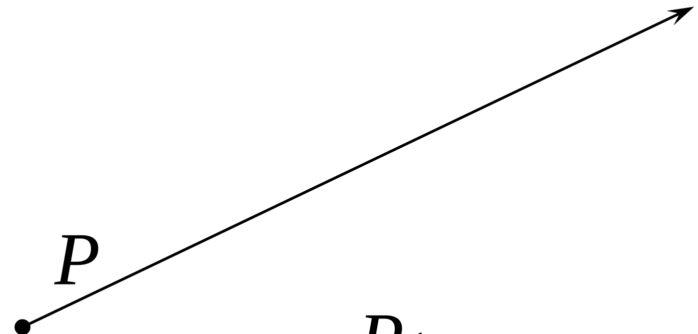
    $L = P_1 + t * (P_2 - P_1)$

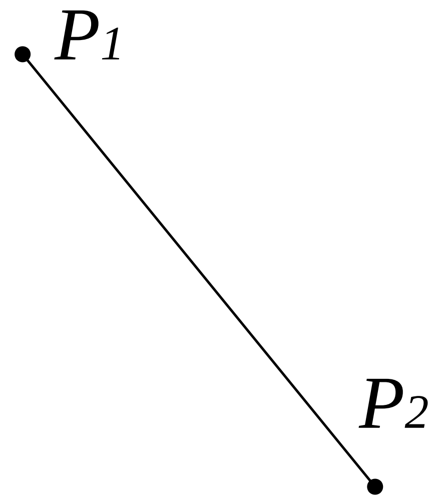


Ray and Segments

- A Ray is a line with

$$0 \leq t \leq \infty$$





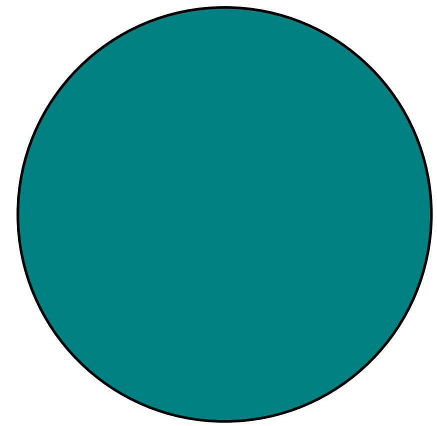
- A segment is a line with

$$0 \leq t \leq 1$$

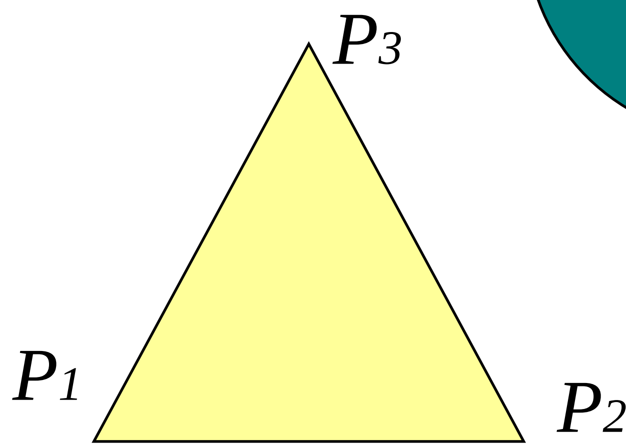
Other Simple Objects

- Circle

$$x^2 + y^2 = R^2$$



- Triangle



Inside Points

Area

Affine Combinations

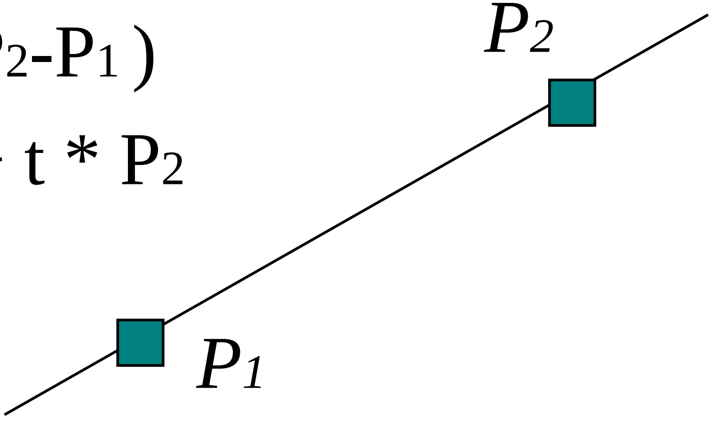
- Affine combination of points P_i is given by

$$P = \sum_i a_i P_i \qquad where, \qquad \sum_i a_i = 1$$

Consider, the equation of line

$$L = P_1 + t * (P_2 - P_1)$$

$$= (1-t) * P_1 + t * P_2$$



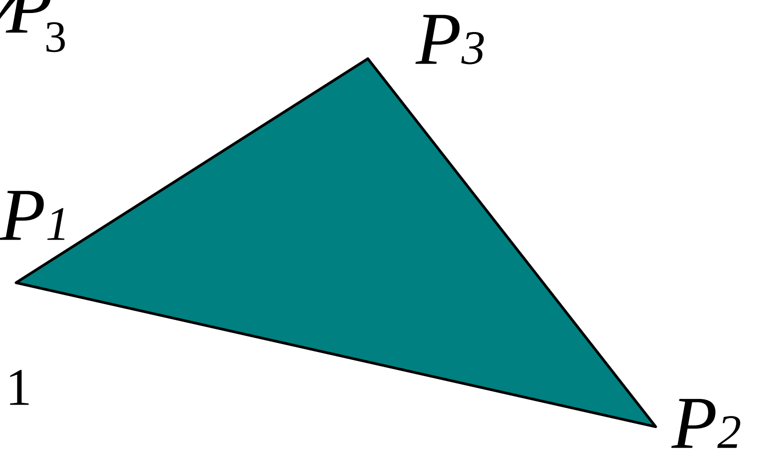
Barycentric Co-ordinates

- Any point on a plane can be written as affine combination of three distinct points

$$P = \alpha P_1 + \beta P_2 + \gamma P_3$$

where,

$$\alpha + \beta + \gamma = 1$$



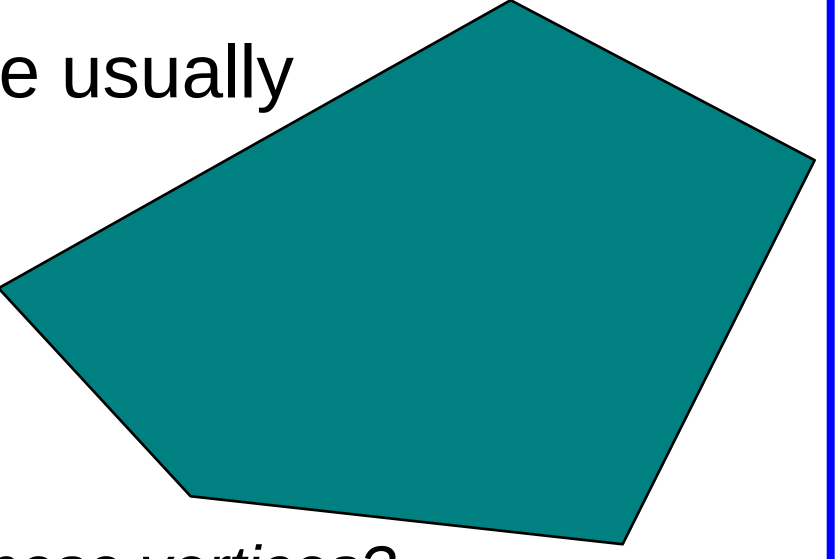
Note that, if $0 < (\alpha, \beta, \gamma) < 1$

point P lies inside the triangle

Polygons

- Polygon is a closed figure formed by an ordered set of points

These corner points are usually termed as *Vertices*

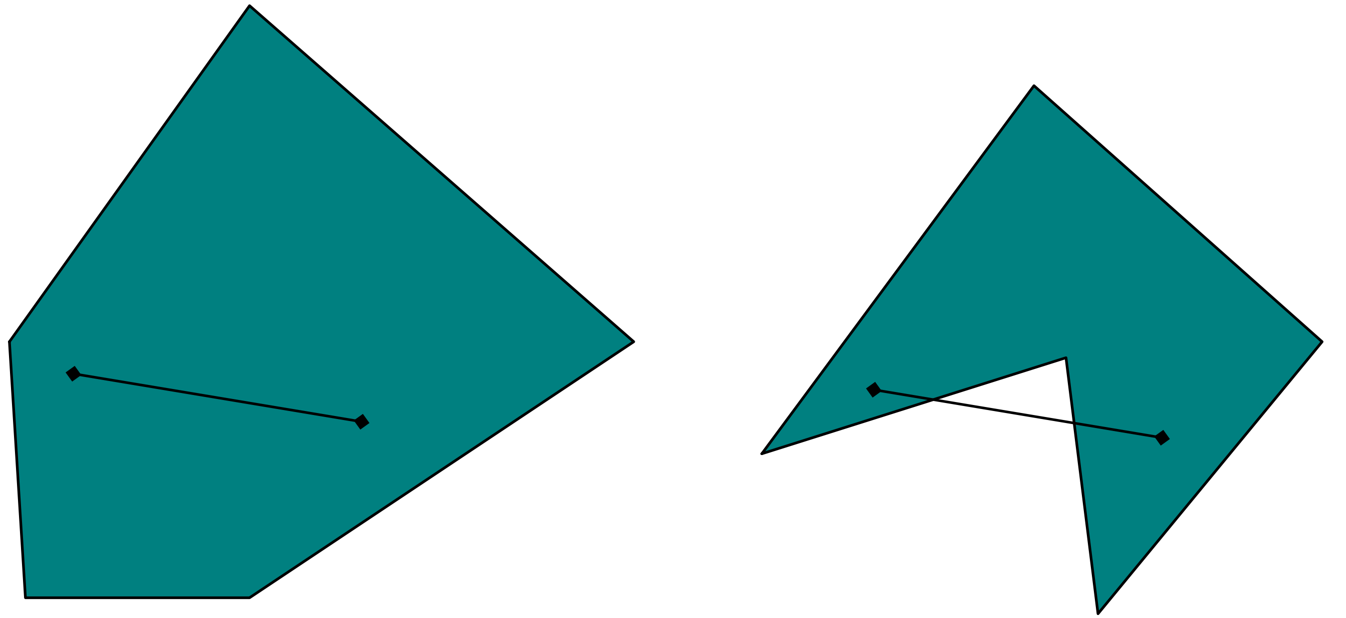


What can be said about the affine combination of these vertices?

Convex Combination

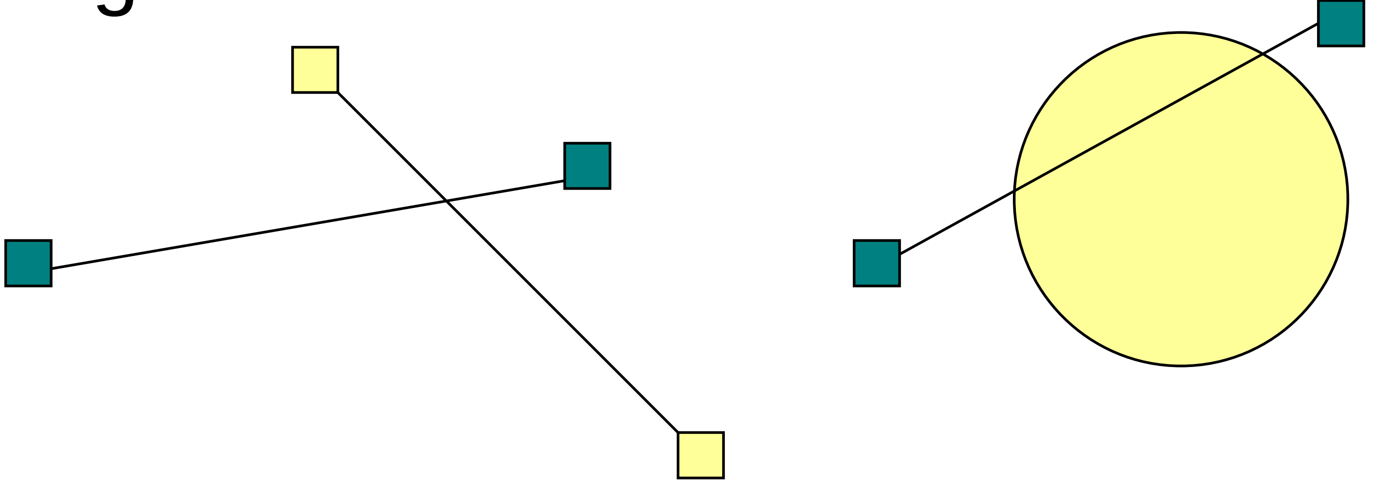
- Convex combination is an affine combination, where all the *weights* are constrained to interval [0,1]
- Given a polygon, if all convex combinations of the vertices of the polygon lie inside the polygon, then we say that, the polygon is *Convex* otherwise, it is said to be *Concave*

Convex and Concave Polygons



Intersections

- Finding intersections between various geometric objects is one of the important task in Computer Graphics Algorithms

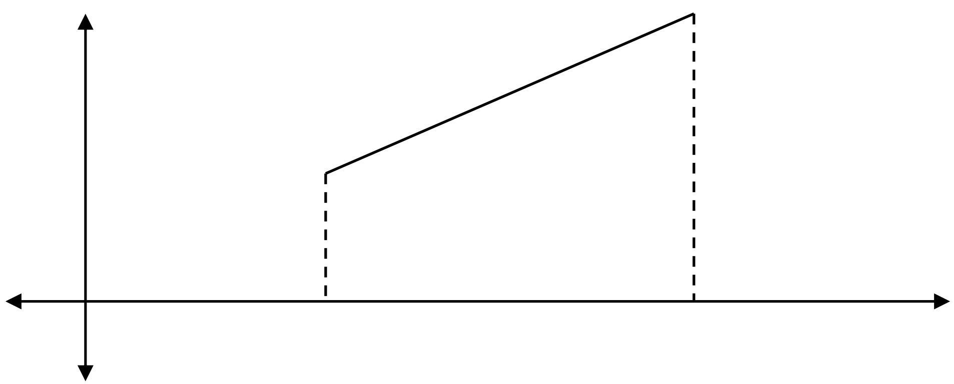


Area Computations

- Finding lengths, areas, volumes, etc.. of various geometric objects is another important problem in Computer Graphics

How to find an area of a polygon in 2D?

Hint:



Matrix

- Two dimensional arrangement of numbers

  In a sense, it is a generalization of vectors

- It has number of rows and columns

$$M_{mn} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \cdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Various Matrices

- Square Matrix

  Matrix whose columns = rows

- Identity Matrix: I

  A square matrix with diagonal elements = 1 and other elements = 0

- Transpose of a Matrix: A^T

Matrix Operations

- Scalar Multiplication

  R[i,j] = k * A[i,j]
- Addition of Matrices

  R[i,j] = A[i,j] + B[i,j]
- Multiplication of Matrices

  R[i,j] = $\sum$ A[i,k] * B[k,j]

  In general, AB $\neq$ BA

Vector - Matrix Multiplication

- Example of Vector - Matrix Multiplication

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \cdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \bullet \begin{bmatrix} x \\ y \\ \vdots \\ w \end{bmatrix} = \begin{bmatrix} a_{11}x + a_{12}y + \ldots + a_{1n}w \\ a_{21}x + a_{22}y + \ldots + a_{2n}w \\ \ldots \\ a_{m1}x + a_{m2}y + \ldots + a_{mn}w \end{bmatrix}$$

Matrix - Matrix Multiplication

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \ldots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \bullet \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \ldots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nn} \end{bmatrix}$$

$$= \begin{bmatrix} a_{11}b_{11}+a_{12}b_{21}+..., & a_{11}b_{12}+a_{12}b_{22}+..., & \cdots & a_{11}b_{1n}+a_{12}b_{2n}+..., \\ a_{21}b_{11}+a_{22}b_{21}+..., & a_{21}b_{12}+a_{22}b_{22}+..., & \cdots & a_{21}b_{1n}+a_{22}b_{2n}+..., \\ \vdots & \ldots & \ddots & \vdots \\ a_{n1}b_{11}+a_{n2}b_{21}+..., & a_{n1}b_{12}+a_{n2}b_{22}+..., & \cdots & a_{n1}b_{1n}+a_{n2}b_{2n}+..., \end{bmatrix}$$

Matrix Inverse

- Inverse of Matrix: A^{-1}

  This is usually used while solving a set of linear equations:

  $Ax = B$

  $x = A^{-1}B$

Thank You